

Day2 Advance SQL Workshop

Problem statements Leetcode:

1] <https://leetcode.com/problems/rank-scores/>

```
SELECT score, DENSE_RANK() OVER(WND) `rank`  
FROM Scores  
WINDOW wnd AS (ORDER BY score DESC);
```

 Code |  Testcase |  Test Result

Accepted Runtime: 130 ms

• Case 1

Input

Scores =

id	score
1	3.5
2	3.65
3	4
4	3.85
5	4
6	3.65

Output

score	rank
4	1
4	1
3.85	2
3.65	3
3.65	3
3.5	4

Expected

score	rank
4	1
4	1
3.85	2
3.65	3
3.65	3
3.5	4

Activate Windows

Go to Settings to activate Windows.

 Contribute a testcase

2] <https://leetcode.com/problems/second-highest-salary/>

Code

```
MySQL Auto
```

```
1 with cte as(  
2 select salary as SecondHighestSalary,  
3 dense_rank() over(w) rnk from Employee  
4 window w as (order by salary)  
5 )  
6 select SecondHighestSalary from cte  
7 where rnk=2;
```

Saved Ln 6, Col 27

Testcase | Test Result

- Case 1
- Case 2

Input

Employee =

id	salary
1	100
2	200
3	300

Output

SecondHighestSalary
200

Expected

SecondHighestSalary
200

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</> Code

MySQL  Auto   

```
1 with cte as(  
2   select salary as SecondHighestSalary,  
3   dense_rank() over(w) rnk from Employee  
4   window w as (order by salary)  
5 )  
6 select max(SecondHighestSalary) as SecondHighestSalary from cte  
7 where rnk=2;
```

Saved

Ln 6, Col

 Testcase |  Test Result

Accepted Runtime: 218 ms

• Case 1

• **Case 2**

Input

Employee =

id	salary
1	100

Output

SecondHighestSalary
null

Expected

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3] <https://leetcode.com/problems/nth-highest-salary>

Code

MySQL Auto

```
1 CREATE FUNCTION getNthHighestSalary(N INT) RETURNS INT
2 BEGIN
3     RETURN (
4         with cte as(
5             select salary,
6                 dense_rank() over(order by salary desc) as rnk from Employee
7         )
8         select max(salary)from cte
9         where rnk=N
10    );
11 END
```

SavedLn 1, Col 1

TestcaseTest Result

Input

Employee =

id	salary
1	100
2	200
3	300

argument =
2

Output

getNthHighestSalary(2)
200

Expected

getNthHighestSalary(2)
200

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4] <https://leetcode.com/problems/department-highest-salary/description/>

RunSubmit🔍📄

Code

MySQLAuto

```
1 SELECT d.name as Department,
2 e.name as Employee,
3 e.salary as Salary
4 FROM (
5 SELECT departmentId, MAX(salary) maxsal FROM Employee
6 GROUP BY departmentId
7 ) AS dm
8 INNER JOIN Employee e ON e.departmentId = dm.departmentId
9 INNER JOIN Department d ON e.departmentId=d.id
10 WHERE e.salary = dm.maxsal;
11
12
```

SavedLn 9, Col 45

TestcaseTest Result

Department =

id	name
1	IT
2	Sales

Output

Department	Employee	Salary
IT	Jim	90000
Sales	Henry	80000
IT	Max	90000

Expected

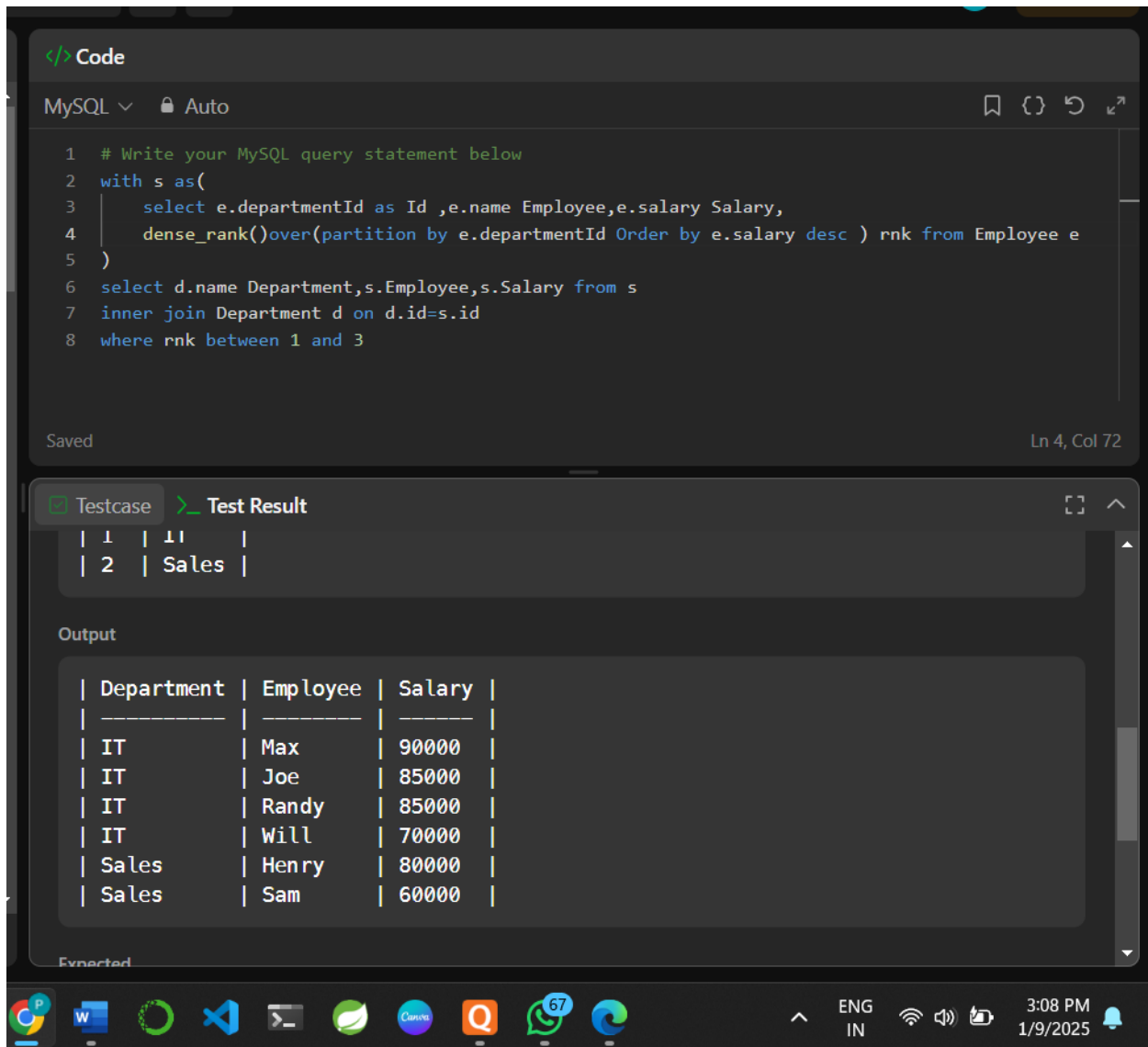
Department	Employee	Salary
IT	Jim	90000
Sales	Henry	80000
IT	Max	90000

Contribute a testcase

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5] <https://leetcode.com/problems/department-top-three-salaries/>



The screenshot shows a code editor with a MySQL query and its test results. The query is as follows:

```
1 # Write your MySQL query statement below
2 with s as(
3     select e.departmentId as Id ,e.name Employee,e.salary Salary,
4     dense_rank()over(partition by e.departmentId Order by e.salary desc ) rnk from Employee e
5 )
6 select d.name Department,s.Employee,s.Salary from s
7 inner join Department d on d.id=s.id
8 where rnk between 1 and 3
```

The test results are as follows:

Department	Employee	Salary
IT	Max	90000
IT	Joe	85000
IT	Randy	85000
IT	Will	70000
Sales	Henry	80000
Sales	Sam	60000

The code editor also shows a "Testcase" tab with the following input:

1	IT
2	Sales

The "Output" tab shows the following result:

Department	Employee	Salary
IT	Max	90000
IT	Joe	85000
IT	Randy	85000
IT	Will	70000
Sales	Henry	80000
Sales	Sam	60000

The code editor also shows a "Test Result" tab with the following output:

Department	Employee	Salary
IT	Max	90000
IT	Joe	85000
IT	Randy	85000
IT	Will	70000
Sales	Henry	80000
Sales	Sam	60000

6] <https://leetcode.com/problems/rising-temperature/>

The screenshot shows a code editor with a MySQL query and its test results. The query is as follows:

```
1 with cte as(
2     select id,recordDate,temperature,
3         LAG(temperature) over(order by recordDate) as prevtemp,
4         DATEDIFF(recordDate,LAG(recordDate) over(order by recordDate)) as diff
5     from Weather
6 )
7 select id Id from cte
8 where temperature > prevtemp and diff=1;
```

The test results show the following output:

Id
2
4

The expected output is the same as the actual output:

Id
2
4

The bottom of the screen shows a Windows taskbar with various application icons and system information: ENG IN, 3:00 PM, 1/9/2025.

7] <https://leetcode.com/problems/consecutive-numbers>

Code

MySQL Auto

```
2 select num,
3 lag(num)over(w) as prev_num,
4 lead(num)over(w)as next_num
5 from Logs
6 window w as(order by id)
7 )
8
9 select distinct num as ConsecutiveNums from cte
10 where num=prev_num and next_num;
11
```

Saved Ln 4, Col 32

Testcase **Test Result**

2	1	
3	1	
4	2	
5	1	
6	2	

View more

Output

ConsecutiveNums
1

Expected

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